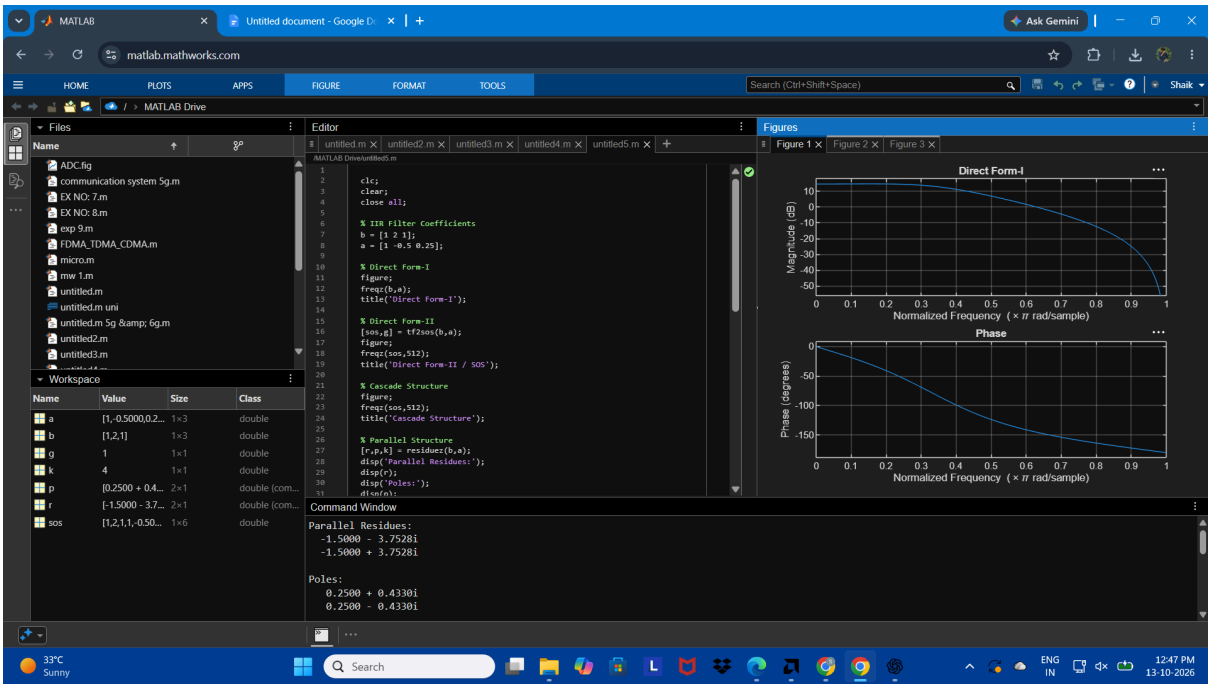
EXP NO:- 5

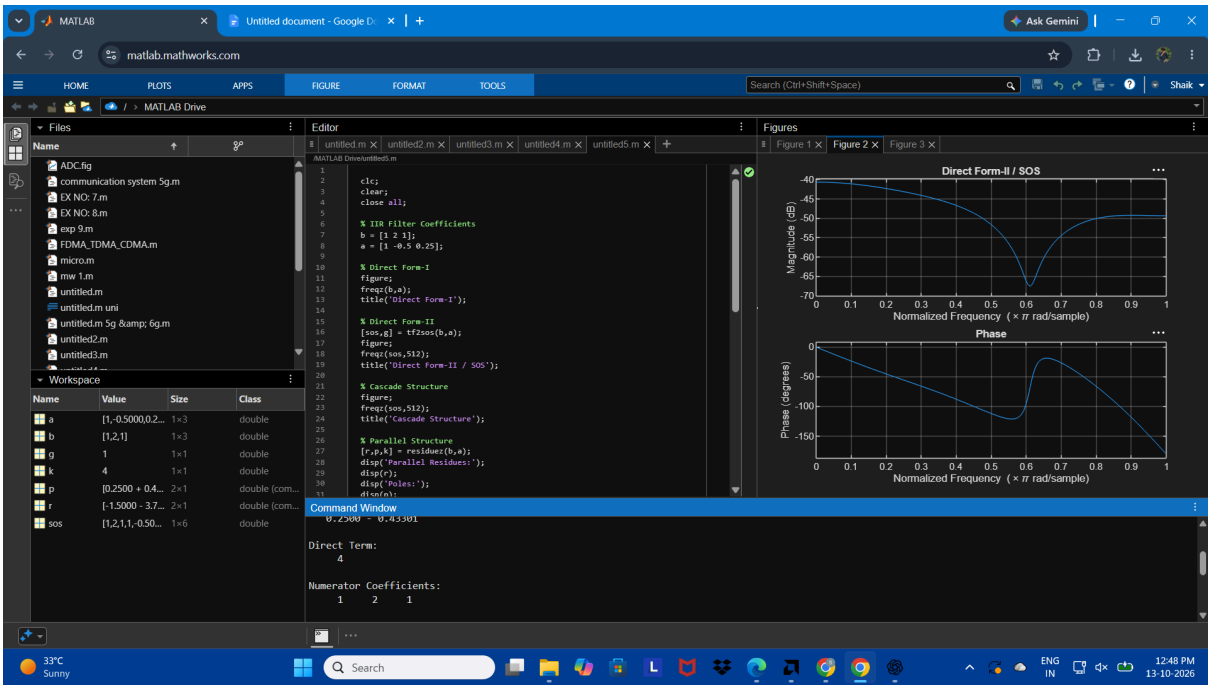
PROGRAM

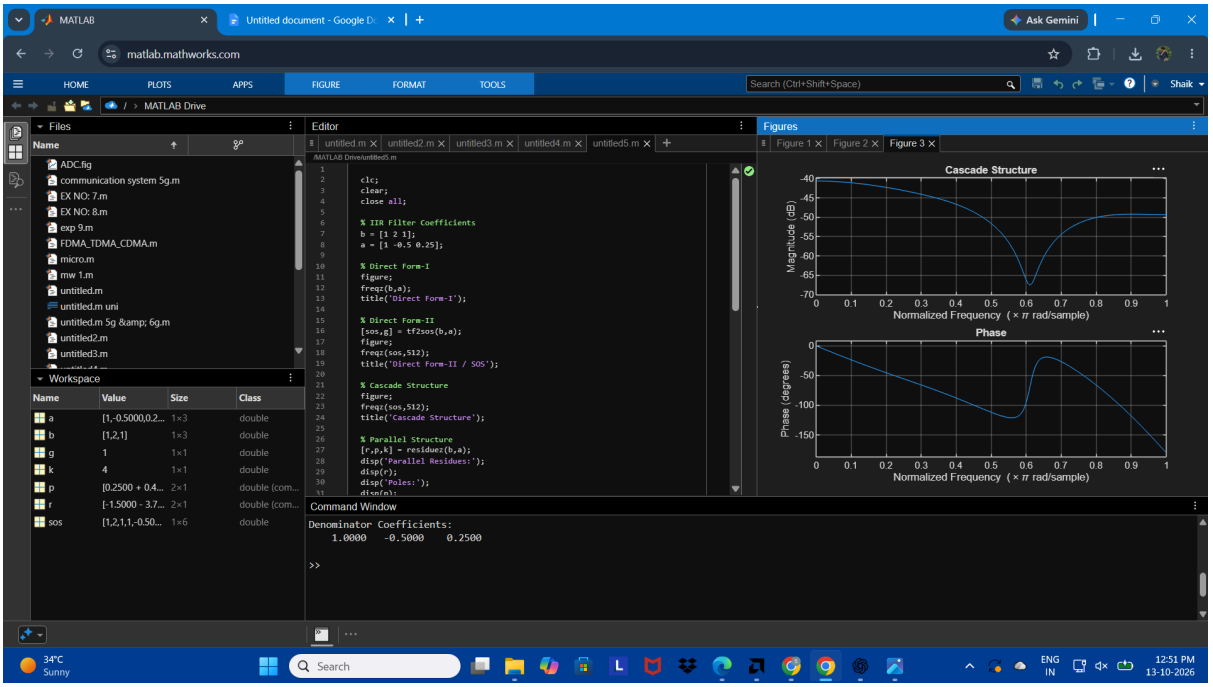
```
clc;
clear;
close all;
% IIR Filter Coefficients
b = [1 2 1];
a = [1 -0.5 0.25];
% Direct Form-I
figure;
freqz(b,a);
title('Direct Form-I');
% Direct Form-II
[sos,g] = tf2sos(b,a);
figure;
freqz(sos,512);
title('Direct Form-II / SOS');
% Cascade Structure
figure;
freqz(sos,512);
title('Cascade Structure');
% Parallel Structure
[r,p,k] = residuez(b,a);
disp('Parallel Residues:');
disp(r);
disp('Poles:');
disp(p);
disp('Direct Term:');
disp(k);
% Display Coefficients
disp('Numerator Coefficients:');
```

```
disp(b);
disp('Denominator Coefficients:');
disp(a);
```

OUTPUT:-






MATLAB
Untitled document - Google D
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Shaik
MATLAB Drive
Files
Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
a
[1,-0.5000,0.2...
1×3
double
b
[1,2,1]
1×3
double
g
1
1×1
double
k
4
1×1
double
p
[0.2500 + 0.4...
2×1
double (com...
r
[-1.5000 - 3.7...
2×1
double (com...
sos
[1,2,1,1,-0.50...
1×6
double
Editor
untitled.m
untitled2.m
untitled3.m
untitled4.m
untitled5.m
MATLAB Drive/untitled5.m
clc;
clear;
close all;
% IIR Filter Coefficients
b = [1 2 1];
a = [1 -0.5 0.25];
% Direct Form-I
figure;
freqz(b,a);
title('Direct Form-I');
% Direct Form-II
[sos,g] = tf2sos(b,a);
figure;
freqz(sos,512);
title('Direct Form-II / SOS');
% Cascade Structure
figure;
freqz(sos,512);
title('Cascade Structure');
% Parallel Structure
[r,p,k] = residuez(b,a);
disp('Parallel Residues:');
disp(r);
disp('Poles:');
Figures
Figure 1
Figure 2
Figure 3
Cascade Structure
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
Denominator Coefficients:
1.0000 -0.5000 0.2500
>>
34°C
Sunny
Search
ENG
IN
12:51 PM
13-10-2026